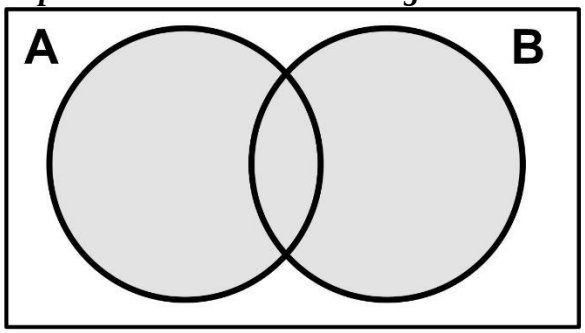
SETS

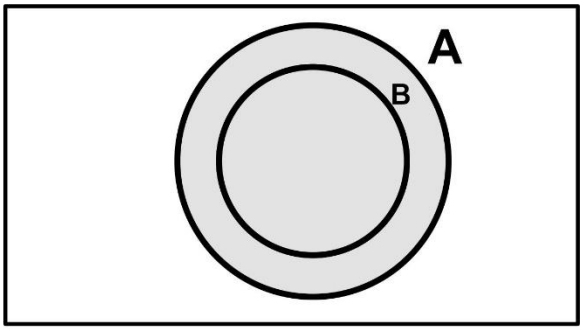
Union and Intersection of Sets

The Union of Sets: The union of two sets say A and B written as $(A \cup B)$ is the set which consists of all the elements that belong to A or to B or to both.
Mathematically written as $A \cup B = \{x \in M \ or \ x \in A \ \ or \ x \in B)$

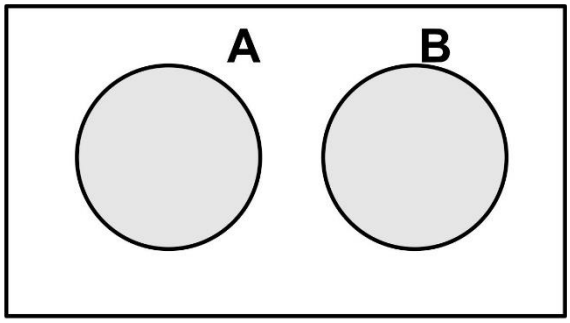
Representation on a Venn Diagram



A∪B

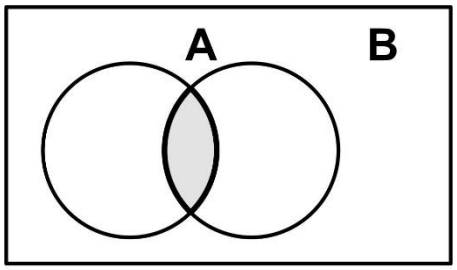


A∪B

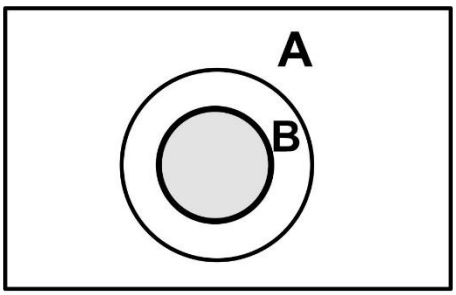


A∪B

Intersection of Sets: The intersection of the sets A and B $(A \cap B)$ is the set which consists of elements that belong to both A and B. Mathematically written as
$A \cap B = \{x \in M \ or \ x \in A \ and \ x \in B\}$



A∩B



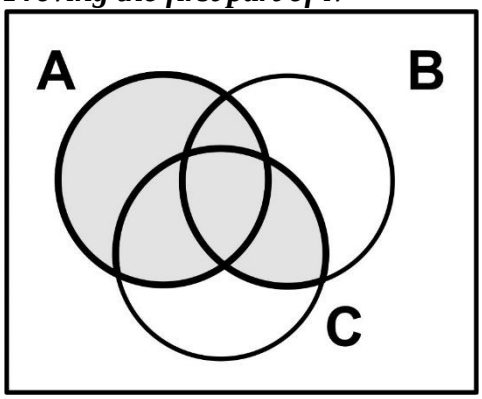
A∩B

Theorem
Let A, B and C be sets.
Then,
i. $A \cup A = A \cap A = A$
ii. $A \cup B = B \cup A$
$A \cap B = B \cap A$

iii. $(A \cup B) \cup C = A \cup (B \cup C)$
$(A \cap B) \cap C = A \cap (B \cap C)$
iv. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

This implies that the union and intersection is always commutative and associate.

Proving the first part of iv



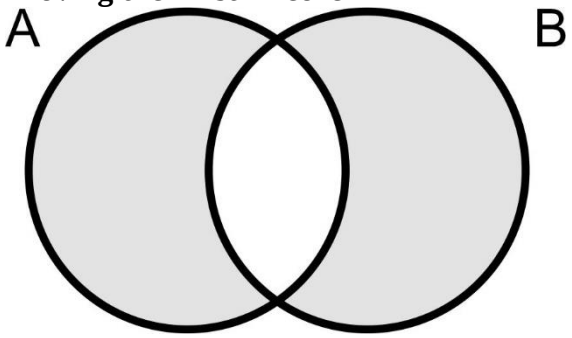
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Using $A \subseteq B \; and \; B \subseteq A \; then \; A = B$
Let $x \in A \cup (B \cap C)$
$\Rightarrow x \in A \; or \; x \in (B \cap C)$
Taking $x \in A$
$\Rightarrow x \in A \; or \; x \in B \; and \; x \in A \; or \; x \in C$
$\Longrightarrow x \in (A \cup B) \; and \; x \in (A \cup C)$
$\Rightarrow x \in (A \cup B) \cap (A \cup C)$
Take $x \in (B \cap C)$
$\Rightarrow x \in B \; and \; x \in C$
$\Rightarrow x \in A \; or \; x \in B \; and \; x \in A \; or \; x \in C$
$\Rightarrow x \in (A \cup B) and \; x \in (A \cup C)$
$x \in (A \cup B) \cap (A \cup C)$
Since $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \; and \; (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$
$\therefore A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Definition: Let A and B be subsets of the universal set (U), the difference of A and B is written as $A - B$ is the set $A - B = \{x: x \in and \; x \notin B\} = A \cap B^{|}$
$U = \{1, 2, 3, 4, 5, 6\}$
$A = \{1, 2, 3\}$
$B = \{3, 4, 5\}$
$A - B = \{1, 2\}$
$B - A = \{4, 5)$

Theorem: De Morgan's Laws
i. $(A \cap B)^{|} = A^{|} \cup B^{|}$
$(A \cup B)^{|} = A^{|} \cap B^{|}$
ii. $A \cap (A \cap B^{|})^{|} = A \cap B$
$A - (A - B) = (A \cap B)$
iii. $A \cap (B \cup C)^{|} = (A \cap B^{|}) \cap (A \cap C^{|})$

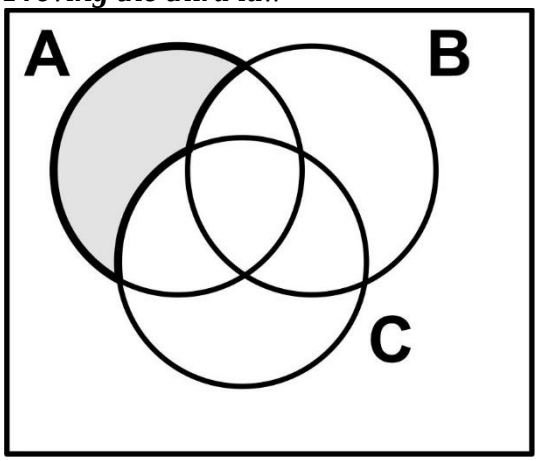
$A - (B \cup C) = (A - B) \cap (A - C)$
iv. $A \cap (B \cap C)^{|} = (A \cap B^{|}) \cup (A \cap C^{|})$
$A - (B \cap C) = (A - B) \cup (A - C)$

Proving the First Theorem



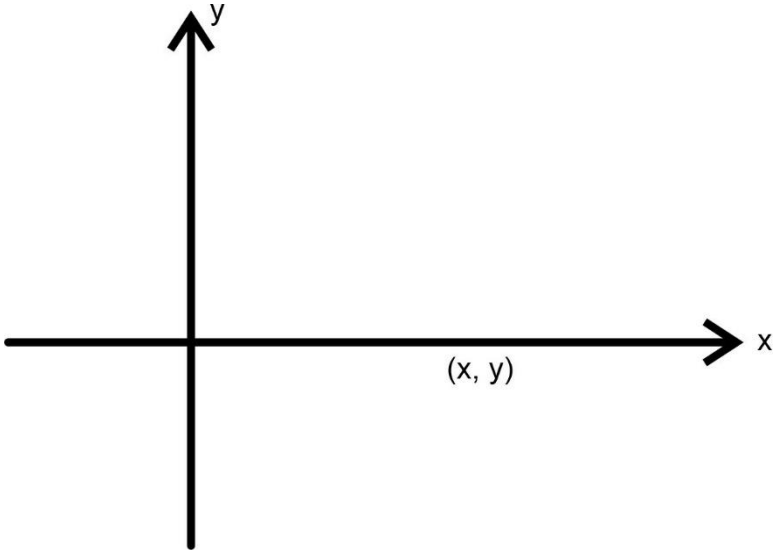
$(A \cap B)^{|}$
Let $x \in (A \cap B)^{|}$
$\Rightarrow x \notin (A \cap B)$
$\Rightarrow x \in A \; and \; x \notin B$
$\Longrightarrow x \in A^{|} \; or \; x \notin B^{|}$
$\Rightarrow x \in A^{|} \; or \; x \notin B^{I}$
$\Rightarrow x \in A^{|} \cup B^{|}$

Proving the third law



Let $x \in A \cap (B \cup C)^{|}$
$\Rightarrow x \in A \; and \; x \notin (B \cup C)$
$\Rightarrow x \in A \; and \; x \; \notin (B \cup C)$
$\Rightarrow x \in A \; and \; x \notin B \; or \; x \notin C$
$\Rightarrow x \in A \; and \; x \in B^{|} or \; x \notin C^{|}$
$\Rightarrow \left(x \in A \; and \; x \; \in B^{|}\right) \; and \; (x \in A \; and \; x \; \in C)$
$\Rightarrow \left(x \in A \; and \; x \in B^{|}\right) \; and \; (x \in A \; and \; x \in)$
$\Rightarrow x \in (A \cap B^{|}) \cap (A \cap C^{|})$

Cartesian Product of Sets



$S \times T = \{s, t\} \in S \times T\}$
The cartesian product of two sets S and T is defined as $S \times T = \{P/P\ = (s, t),\ s \in S, t \in T)$
That is, the set of all ordered pairs of elements (s, t) such that the first member of each pair always belongs to S and the second member always belong to T.

$S = \{2, 4\}\ T = \{2, 4, 7\}$
$S \times T = \{2{,}2)(2, 4), (2, 7), (4, 2), (4, 4), (4, 7)\}$
Some properties of Cartesian Product of Sets
i. Two elements of S × T are equal if and only if they are identical.
i.e. (s, t) = (a, b) ⇒ $S = a\ and\ t = b$

2. If $s = \emptyset\ \ or\ T = \emptyset\ then\ S \times T = \emptyset$
3. $S \times T \neq T \times S$
4. The cartesian product of sets is not associative.

Prove that A × B = B × A if and only if either A – B or at least one set is empty.
Proof
Let A = B, A × B = B ×
B × B = B × B = B^2
A × B = $\emptyset, = B \times A$
Assuming that $A = \ \emptyset, B = \ \emptyset$
Let A × B = B × A
And also let (a, b) ∈ A × B
⇒ a ∈ A and b ∈ B
(b, a) ∈ B × A
⇒b ∈ B and a ∈ A
Since A ⊆ B and B ⊆ A
∴ A = B

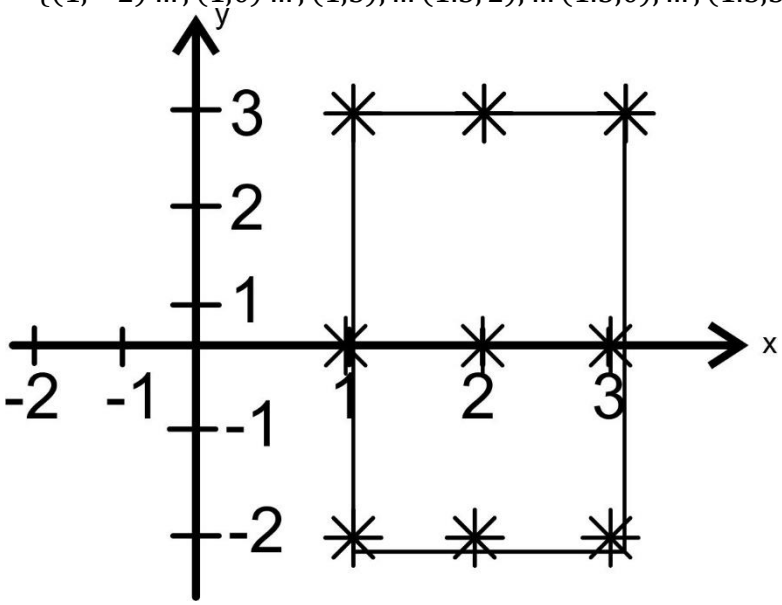
A × (B × C) ≠ $(A \times B) \times$
Prove that
A × (B × C) = (A × B) ×
If and only if at least one of the sets A, B, C is empty.

Proof
Assume A = ∅, B = ∅ or C = ∅
A × (B × C) = ∅ = (A × B × C)
Let (*x*, y) ∈ A × (B ×) and assume A ≠ ∅, B ≠ ∅ and C ≠ ∅
$\Rightarrow x \in A \; and \; y \; \in (B \times)$
And let (*x*, *y*) ∈ $(A \times B) \times C$
$\Rightarrow x \in (A \times B) and \; y \in C$
But if $x \in A, x \notin (A \times B) \; and \; y \; \in (B \times C) y \notin C$ and this is a contradiction to the earlier statement.
∴ A, B, C are empty or at least one is empty.

Let A = {1, 2}, B = {3, 4} and C = {5, 6}
Find A × B × C = $\{(1,3,5),(1,3,6),(1,4,5),(1,4,6),(2,3,5),(2,3,6),(2,4,5),(2,4,6)\}$

Example 2: Let $A = \{1 \leq x \leq 2\}$
$B = \{-2 \leq x \leq 3\}$
Sketch and describe the graph of A × B in the XY plane

Solution
$A \times B$
$= \{(1,-2) \ldots, (1,0) \ldots, (1,3), \ldots (1.5,2), \ldots (1.5,0), \ldots, (1.5,3), \ldots (2,-2), \ldots (2,0) \ldots, (2,3)\}$



The graph is a rectangular figure with vertices,
(1, -2), (3, -2), (3, 3) and (1, 3)

This was solved in class but solve it and see if you get the concept
And also try to prove the rest of the De Morgan's Laws
If $A = \{x \leq 2\} \; and \; B = \{-1 \leq x \leq 3\}$

Find A × B and hence sketch a graph of A × B and describe the type of graph drawn.